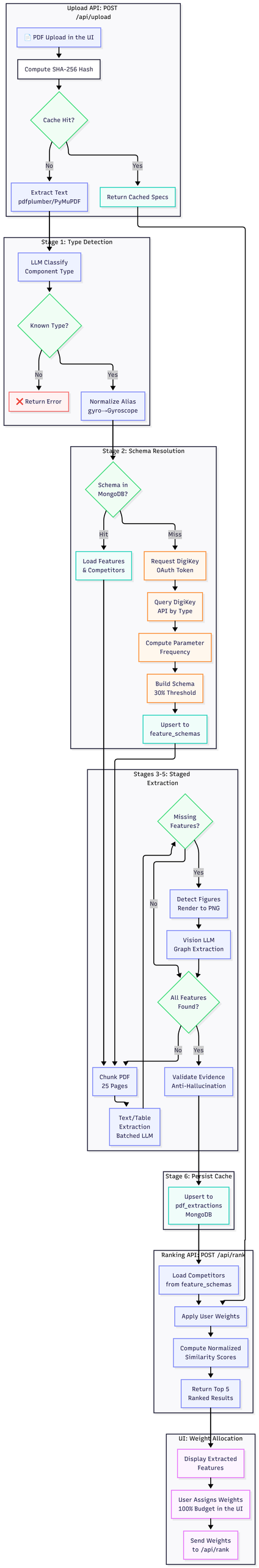
Upload API: POST /api/upload
PDF Upload in the UI
Compute SHA-256 Hash
Cache Hit?
No
Yes
Extract Text pdfplumber/PyMuPDF
Return Cached Specs
Stage 1: Type Detection
LLM Classify Component Type
Known Type?
No
Yes
Return Error
Normalize Alias gyro→Gyroscope
Stage 2: Schema Resolution
Schema in MongoDB?
Hit
Miss
Load Features & Competitors
Request DigiKey OAuth Token
Query DigiKey API by Type
Compute Parameter Frequency
Build Schema 30% Threshold
Upsert to feature_schemas
Stages 3-5: Staged Extraction
Missing Features?
Yes
No
Detect Figures Render to PNG
Vision LLM Graph Extraction
All Features Found?
No
Yes
Chunk PDF 25 Pages
Validate Evidence Anti-Hallucination
Text/Table Extraction Batched LLM
Stage 6: Persist Cache
Upsert to pdf_extractions MongoDB
Ranking API: POST /api/rank
Load Competitors from feature_schemas
Apply User Weights
Compute Normalized Similarity Scores
Return Top 5 Ranked Results
UI: Weight Allocation
Display Extracted Features
User Assigns Weights 100% Budget in the UI
Send Weights to /api/rank